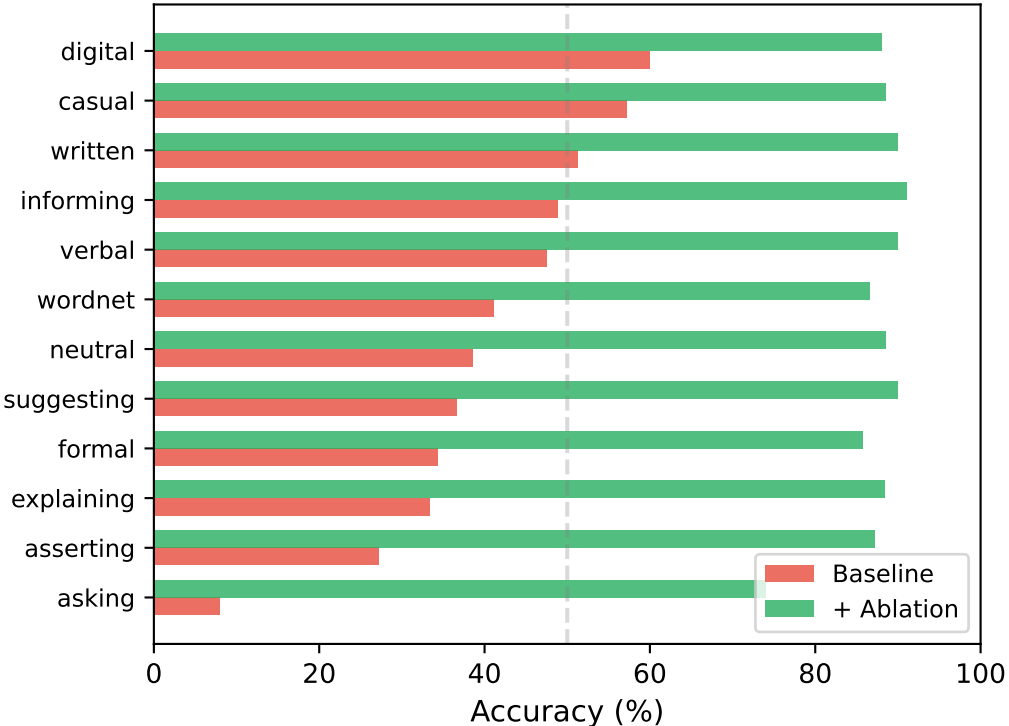
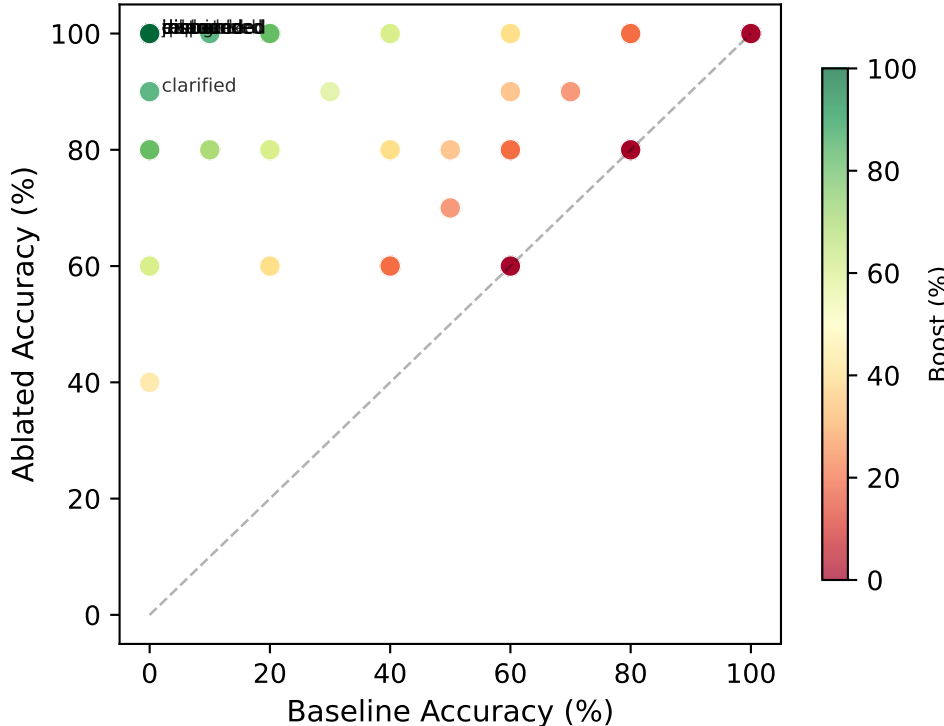


Theory of Mind Circuit Analysis: Massive Linguistic Sweep
Qwen3-4B | 5-Head Ablation (L17H4, L18H11, L18H14, L19H30, L21H17)

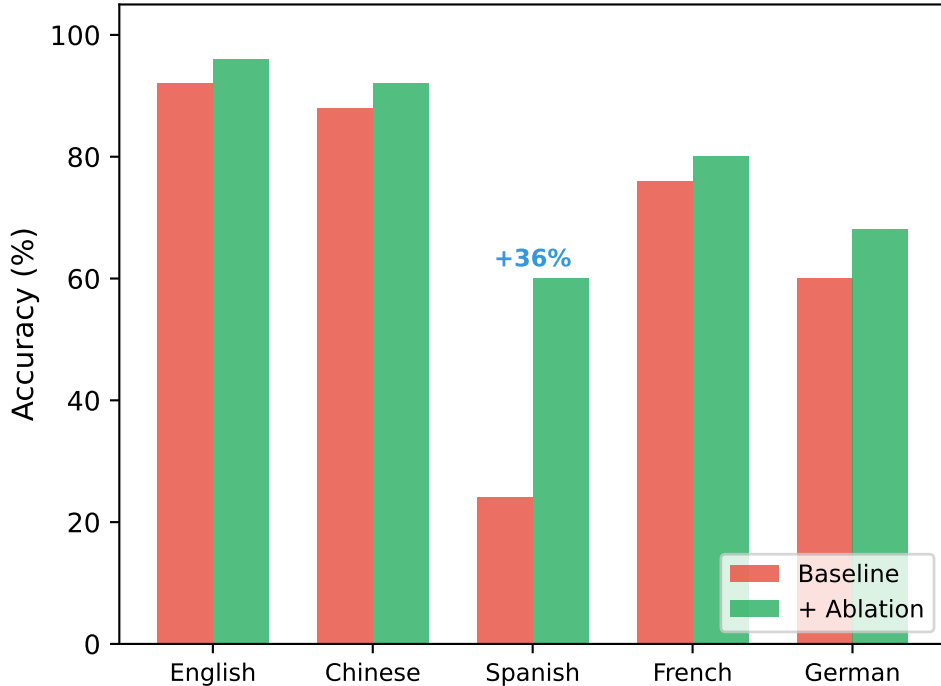
A. ToM by Verb Category



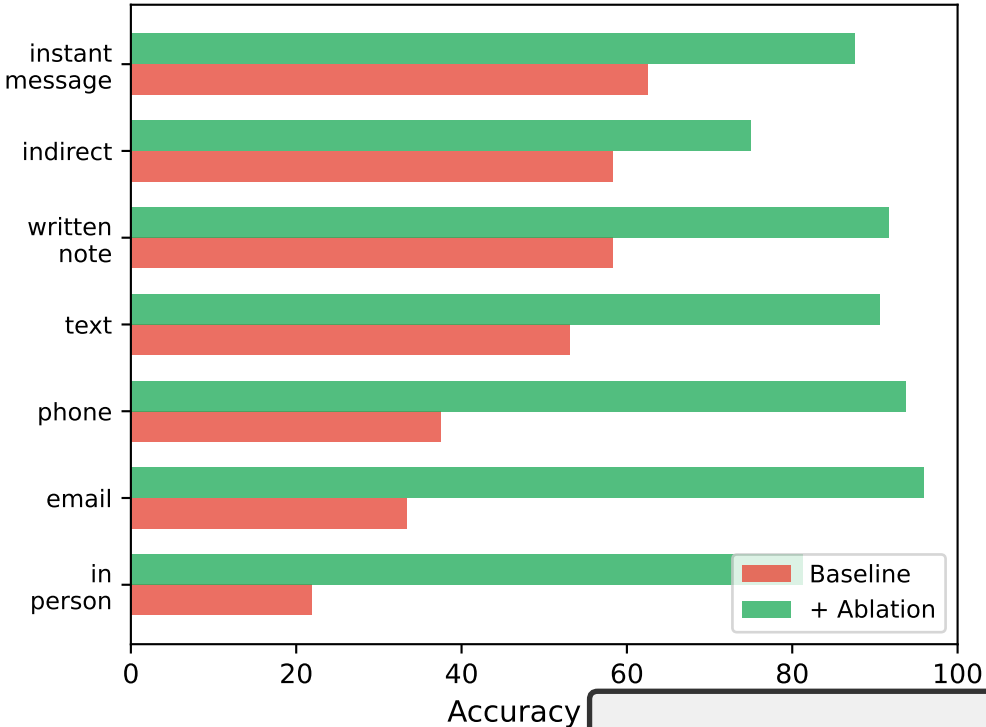
B. 178 Verbs: Baseline vs Ablated



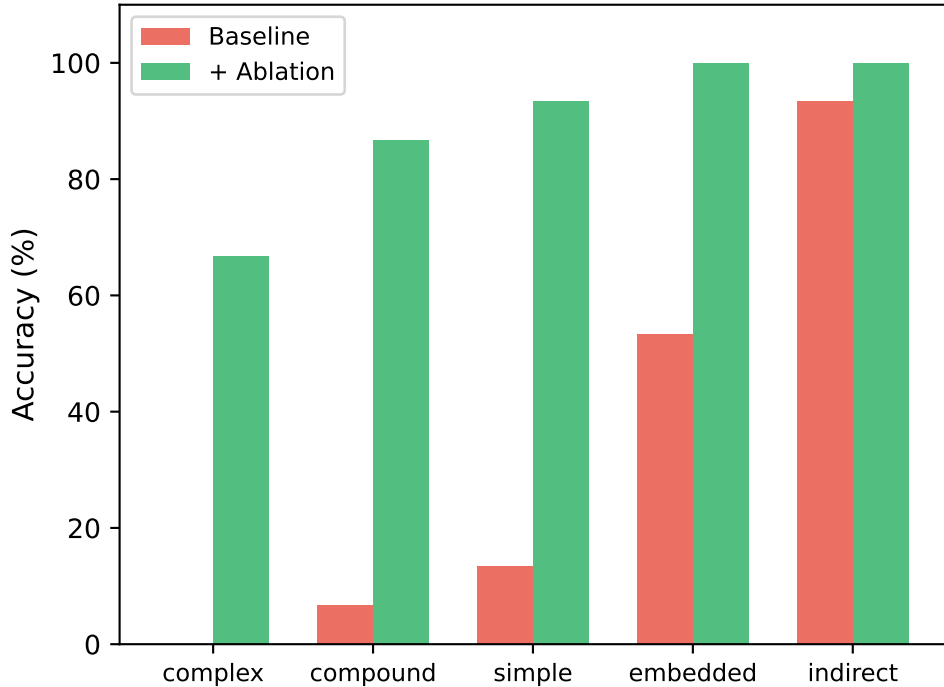
C. ToM Across Languages



D. ToM by Communication Medium



E. ToM by Sentence Structure



F. Best & Worst Verbs

VERB	BASELINE	ABLATED	BOOST
WORST (Inhibition Triggered)			
told	0%	100%	+100%
noted	0%	80%	+80%
clarified	0%	90%	+90%
announced	0%	100%	+100%
indicated	0%	60%	+60%
asked	0%	40%	+40%
requested	0%	60%	+60%
queried	0%	100%	+100%
BEST (No Inhibition)			
manifested	80%	100%	+20%
worded	80%	80%	+0%
dispatched	100%	100%	+0%
provided	100%	100%	+0%
supported	100%	100%	+0%

MASSIVE LINGUISTIC SWEEP SUMMARY

SCALE

- 178 unique verbs tested
- 12 semantic categories
- 5 languages (EN/ZH/ES/FR/DE)
- 7 communication mediums
- 5 sentence structures
- 1,300+ total scenarios

KEY FINDINGS

- Mean baseline: 41.1%
- Mean ablated: 87.1%
- Mean boost: +46.0%
- 23 verbs at 0% baseline (complete failure)
- 83 verbs reach 100% with ablation

SCIENTIFIC INSIGHT

The inhibitory circuit (L17H4, L18H11, L18H14, L19H30, L21H17) is VERB-SENSITIVE:

- DIRECT verbs ("told", "informed") → MAXIMUM inhibition → 0% baseline
- INDIRECT verbs ("provided", "supported") → NO inhibition → 100% baseline

This explains why models fail Sally-Anne tests but succeed at multi-agent collaboration:
Different verb phrasings trigger different levels of the belief-update inhibition circuit.